

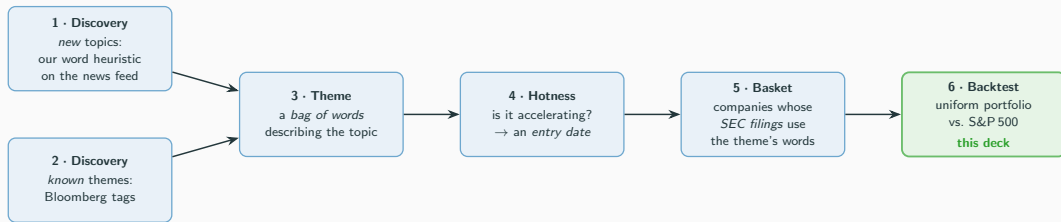
Theme Detection — updates VIII

A first baseline, bag-of-words only: from detection to a backtested basket

August 2026

The chain now runs end to end. Everything in this deck uses **words alone** — no prices, no fundamentals, no tuning — so it is the **floor** every smarter version must beat. Two worked examples (quantum, genAI) against the S&P 500.

The pipeline — last stage is new (1/10)



What this baseline is. Words all the way through: words discover the theme, words date its acceleration, words (counted in SEC filings) pick the companies. The portfolio is uniform and the backtest has no free parameters. Deliberately the simplest thing that could work — so the two examples measure the *method*, not our tuning.

Detection: two routes to a theme, one hotness score (2/10)

New themes — the word-novelty heuristic. A theme being born mints new vocabulary. We keep the words never seen in a two-year baseline and promote those that cluster, persist and bridge across headlines:

candidates → caught → promoted

genAI: *chatgpt* caught in the week of Jan 3 2023, promoted Jan 17 — seven weeks after launch, four months before the first genAI ETF.

Known themes — Bloomberg tags. If a topic code already exists, the tag *is* the theme definition and we skip discovery entirely.

Hotness

Either way, the theme's temperature is the rolling z-score of its weekly headline count,

$$z_t = \frac{c_t - \mu_{t-1}}{\sigma_{t-1}},$$

which ranks competing themes and gives the **entry date** for the backtest.

Why both routes

The heuristic sees themes *before* any tag exists; tags cover mature themes at zero cost.

Portfolio: uniform weights, on purpose (3/10)

Basket. A company enters if its annual/quarterly SEC reports mention the theme's words (point-in-time: only filings available at the entry date). Rank by mention counts, keep the top names. The rule can also refuse: **at the 2018 quantum-ETF date it selects zero companies — no investable theme, no trade.**

Weights. At every rebalance date each of the N held names gets

$$w_j = \frac{1}{N} \quad (\text{equal euros per name}).$$

Uniform needs no estimates and no optimizer — the right property for a floor. Note that resetting to $1/N$ each quarter means **selling the quarter's winners to buy its losers**: a real economic choice, whose cost we measure on slide 7.

Prices. Split- and dividend-adjusted daily closes, so price ratios always compare comparable per-share values (verified on NVDA's 2024 10-for-1 split: no discontinuity).

Backtest: four formulas, no knobs (4/10)

Buy at the theme's entry date, hold m months, rebalance, repeat. Periods are **contiguous**: each exit day *is* the next entry day — no return falls between.

Time runs on a **grid of rebalance days**, not continuously: t_i = first trading day $\geq$ entry + $i \cdot m$ months, for $i = 0, \dots, n$, with t_n = end date. Period i goes from t_i to t_{i+1} and its return uses *only those two endpoints*, equal-weighted over the held names H_i (basket members with a price at t_i):

$$r_i = \frac{1}{|H_i|} \sum_{j \in H_i} \left(\frac{P_j(t_{i+1})}{P_j(t_i)} - 1 \right), \quad i = 0, \dots, n-1$$

$$R = \sum_i r_i, \quad \sigma = \text{std}(r_i), \quad \text{ratio} = \frac{R}{\sigma},$$

$$\text{Sharpe}_{\text{ann}} = \frac{\bar{r}}{\sigma} \sqrt{12/m},$$

where m is the rebalance interval in months, so $12/m$ = periods per year (here $m = 3$: $\sqrt{12/3} = 2$).

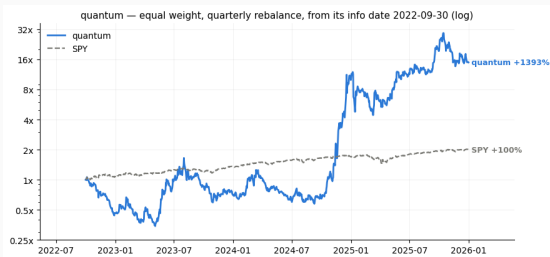
Which ratio when

R/σ grows with the number of periods: fair only *within* one window. The annualized Sharpe is horizon-free — it is how we compare experiments with different entry dates.

Benchmark

The S&P 500 (dividend-adjusted, via SPY) runs through the *same* formula on the *same* days — apples to apples by construction.

Example 1 — quantum: detected early, paid late (5/10)



Entry **2022-09-30**; the rule's four pure-plays (IONQ, QUBT, RGTI, QBTS); 14 quarters.

	sum R	σ	R/σ	Sharpe
quantum	+1670%	406%	4.1	0.59
S&P 500	+72%	5.2%	14.0	2.00
quantum ETF	+126%	11.2%	—	1.62

First quarter after entry: **-54%**; a year below half the starting capital. **91% of the sum is one quarter (Q4-2024)**. The four names move together (correlation **0.93**) — effectively *one* bet, so the basket carries full single-stock volatility. The diversified theme ETF shows what breadth alone buys: 1.62.

3M returns (%) — quantum: -54, -13, +140, -11, -19, +49, -42, +13, **+1516**, -45, +94, +65, -22, -1 • S&P 500: +8, +6, +10, -3, +11, +11, +5, +6, +3, -5, +11, +8, +3, -1

Example 2 — genAI: detected on time, mapped to the wrong companies (6/10)

TODO: Filtro sui ticker —> Rimozione ticker

genai — equal weight, quarterly rebalance, from its info date 2023-01-17 (log)

Parametri: 1. Market cap > 1/10 market cap minima per essere nel Russel

mettete limite tipo 250 milioni USD di floating nella market cap.

Se non c'è quel dato su yahoo, usate 500 come mkt cap totale



Entry **2023-01-17** (the promotion week);
point-in-time top-10 from SEC filings (NVDA, TRIP, GOOG, EA, MYPS, APPS, EXPE, EBAY, CRNC, FFAI); 12 quarters.

	sum R	σ	R/σ	Sharpe
genAI	+101%	26%	3.9	0.64
S&P 500	+62%	7.6%	8.2	1.36

The theme was **seven weeks old**: no filing mentioned its vocabulary yet, so word counts picked pre-ChatGPT “AI-talkers” — travel portals, e-commerce, and the meme stock behind the orange spike. NVDA alone is 101pp of the basket’s 133pp; the other nine add volatility, not theme.

3M returns (%) — genAI: -1, +15, -25, -2, -10, +67, -6, +30, -14, +43, +8, -3 • S&P 500: +5, +9, -3, +9, +6, +12, +5, +3, -12, +20, +6, +3

Results: high return, higher volatility (7/10)

Both baskets beat the S&P 500 by a wide margin on raw return — and both lose to it once risk is counted, in a window where the index's own Sharpe sits at the **90th percentile** of its history (median 3-year Sharpe since 1994: 1.0).

Where the risk comes from

Quantum: concentration — four names, correlation 0.93, one bet at 400%/quarter volatility.

genAI: two lottery tickets dominate the volatility budget, and quarterly rebalancing keeps re-buying them after each collapse — that alone costs **0.33 of Sharpe** vs. buy-and-hold of the same list (0.97).

How fragile

Drop each basket's best quarter: Sharpe 0.59 → 0.40 (quantum), 0.64 → 0.32 (genAI). The index barely moves. One quarter carries everything.

The words were right — in the news

The co-mention basket (slide 10) beats the index (1.94 vs. 1.37): the *mapping*, not the words, was genAI's problem.

Limits (8/10)

1. **Only positive themes so far.** Quantum and genAI are themes we know exploded — selection bias by construction. How often the pipeline buys a theme that *dies* is unmeasured, and it is the question that matters.
2. **Words are not sufficient on SEC filings for newborn themes.** Filings lag the news by months: at genAI's entry date its vocabulary existed in no filing, so counts selected noise. The same words applied to the news worked (Sharpe 1.94–1.99). Filings remain valid for mature themes — quantum's four pure-plays were found without naming them.
3. **No exit rule yet.** Detection gives an *entry* date; nothing tells us when the theme is over, so the backtest simply holds to the end of the window. Quantum shows what is at stake: sell after the crash and the theme is a disaster, hold through it and it is the best trade in the deck. The natural candidate — hotness decay as an exit signal — is untested.
4. **Smaller prints.** No transaction costs or market impact (microcaps!); early quantum quarters trade pre-merger SPAC prices; every Sharpe here rests on one quarter and is a fragile point estimate.

Next: the backtest becomes the judge (9/10)

Stop asking “does one basket beat the index” — run the whole chain over *many* themes and dates, including the duds, and let the outcome label each theme:

$$\text{theme} \mapsto (\text{Sharpe}, R, \sigma, \text{drawdown}) \mapsto \{\text{good}, \text{bad}\}$$

What the labels buy us

The real question becomes answerable: *which detection-time features predict a good theme?* Hotness, novelty type, basket purity, concentration, breadth of the co-mention graph — a classifier trained on our own backtests.

Supporting fixes

Map newborn themes through the news co-mention graph (filings only for mature ones); add simple risk caps to the uniform baseline; keep negative-control themes in the set to keep the classifier honest.

genAI, take two: the detection's co-mentions as the basket (10/10)

Each week the funnel also lists the words *co-mentioned* with the anchor — from the genAI run, in order of appearance:

week (2023)	new co-mentions of <i>chatgpt</i>
Jan 3–9	microsoft
Jan 17–23 (<i>promotion</i>)	openai
Jan 24–30	nvidia , altman
Jan 31–Feb 6	google (Bard)
Feb 7–13	baidu, alibaba, netease (Ernie)
Feb 21–27	amazon (Hugging Face)

Listed ones only, mapped to tickers: **core** = MSFT, NVDA, GOOGL (enter Feb 1) • **ecosystem** = +BIDU, BABA, NTES, AMZN (Mar 1). **No SEC filings involved** — vs. example 2 only the word→company *mapping* changed.

3M returns (%) — news-core: +22, +31, −3, +26, +15, +14, +8, +1, −7, +32, +23, −2 • S&P 500: +2, +10, −7, +16, +3, +9, +5, +5, −6, +12, +10, +0

	sum R	σ	Sharpe
core (3)	+160%	13.8%	1.94
S&P 500 (same window)	+59%	7.1%	1.37
ecosystem (7)	+106%	8.8%	1.99
S&P 500 (same window)	+61%	4.3%	2.37

